

Lecture 1 (2.10.2025) → lecture 0 on Teams

Boolean function - a formula that works only with T/F values.

AND ($\wedge$) → multiplication: $1 \wedge 1 = 1$

OR ($\vee$) → addition: $1 \vee 0 = 1$, $0 \vee 0 = 0$.

NOT ($\neg$) → inverse: $\neg 1 = 0$, $\neg 0 = 1$

→ Boolean algebra

Regular conversion from Any Base → Base 10

$$4312_{(5)} = 2 \cdot 5^0 + 1 \cdot 5^1 + 3 \cdot 5^2 + 4 \cdot 5^3 = \dots (10)$$

Conversion of fractionals Any Base → Base 10

$$43,12_{(5)} = 2 \cdot 5^{-2} + 1 \cdot 5^{-1} + 3 \cdot 5^0 + 4 \cdot 5^1 = 23,28 (10)$$

$$256_{(10)} \begin{array}{r} 16 \\ \hline 165 \\ \vdots \\ 13 \\ \hline 16 \\ \hline 10 \\ \vdots \\ 5 \\ \hline 16 \\ \hline 0 \end{array} \Rightarrow 256_{(10)} = 0A5D_{(16)}$$

Diagram showing the conversion of 256 from base 10 to base 16. The number 256 is divided by 16, yielding a quotient of 16 and a remainder of 0. The quotient 16 is then divided by 16, yielding a quotient of 1 and a remainder of 0. The quotient 1 is then divided by 16, yielding a quotient of 0 and a remainder of 1. The remainders, read from bottom to top, are 0, 5, D, and 0, which correspond to the hexadecimal digits 0, A, 5, and D respectively. A curved arrow indicates the reading of the remainders from bottom to top.

Conversion of a fractional → Any Base

Ex: $543,17 = ?_{(5)}$

$$\begin{array}{l} 543 : 5 = 108 \text{ r } 3 \\ 108 : 5 = 21 \text{ r } 3 \\ \vdots \\ \vdots \end{array}$$

$$\Rightarrow 543 = 4133$$

$$\begin{array}{l} 0,17 \cdot 5 = 0,85 \\ 0,85 \cdot 5 = 4,25 \\ \vdots \\ \vdots \end{array}$$

periodicity

$$\Rightarrow 0,17 = 0,04(1)$$

$$\Rightarrow 543,17 = 4133,04(1)_{(5)}$$

if:
 - periodicity
 - zero
 - no process
 } we STOP.

we don't have an exact representation of that nr.

$$7(8) + 1(8) = 10(8)$$

$$3(10) + 1(10) = 4(10)$$

$$11(2) + 1(2) = 100(2)$$

$$a) 3201_{(4)} = ?_{(2)}$$

$\downarrow$
 $B_4 \Rightarrow$ each B_4 "digit" has 2 B_2 "digits" ($4 = 2^2$)

$$3201_{(4)} = 11 \ 10 \ 00 \ 01_{(2)}$$

$$b) 7205,346_{(8)} =$$

$\downarrow$
 $B_8 \Rightarrow$ each B_8 "digit" has 3 B_2 "digits" ($8 = 2^3$)

$$7205,346_{(8)} = 111 \ 010 \ 000 \ 101,011 \ 100 \ 110_{(2)}$$

Pattern:

$B_2 \rightarrow B_4 \Rightarrow$ groups of 2 ($2^2 = 4$)

ت ت ت ...

$B_2 \rightarrow B_8 \Rightarrow$ groups of 3 ($2^3 = 8$)

تتت تتت تتت

$B_2 \rightarrow B_{16} \Rightarrow$ groups of 4 ($2^4 = 16$)

Write $11(10) = B(16)$ as a sum of pw of 2:

$$B(16) = 11(10) = 1011(2) = 2^3 + 2^1 + 2^0$$

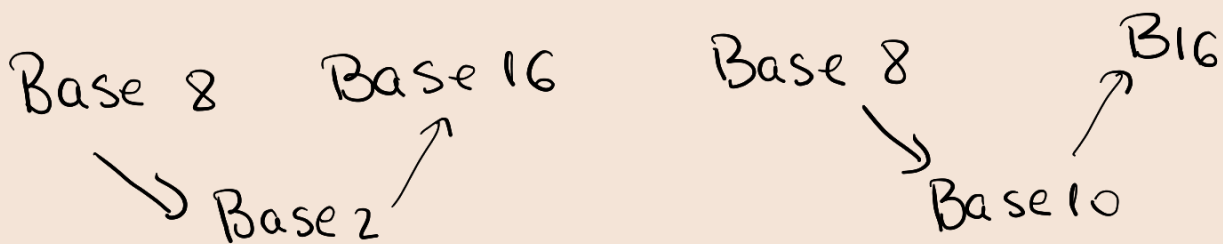
Each Natural number is a sum of powers of 2.

Base 2 $\rightarrow$ Base 2^m

$$\underline{101110}(2) \xrightarrow{\text{Pattern}} \begin{array}{cc} \underline{101} & \underline{110} \\ \downarrow & \downarrow \\ 5 & 6 \end{array} (2) = 56(8)$$

$$\underline{110101110}(2) \xrightarrow{\text{Pattern}} \begin{array}{ccc} \underline{0001} & \underline{1010} & \underline{1110} \\ \downarrow & \downarrow & \downarrow \\ 1 & A & E \end{array} (2) = 1AE(16)$$

$$\underline{110110}(2) \xrightarrow{\text{Pattern}} \begin{array}{ccc} \underline{11} & \underline{01} & \underline{10} \\ \downarrow & \downarrow & \downarrow \\ 3 & 1 & 2 \end{array} (2) = 312(4)$$



X not in the exam!

Read methods of conversions.